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Studierichting: Informatica
Oefeningenreeks 4A nr. 33.1

$$\int e^x \sqrt{1+e^x} dx$$

$$t = 1 + e^x (\Rightarrow dt = e^x dx)$$

$$\int \sqrt{t} dt$$

$$= \int t^{\frac{1}{2}} dt$$

$$= \frac{t^{\frac{3}{2}}}{\frac{3}{2}} + C$$

$$= \frac{2}{3} t^{\frac{3}{2}} + C$$

$$1 + e^x = t$$

$$= \frac{2}{3} (1 + e^x)^{\frac{3}{2}} + C$$

References